Stereo Vision System for Indoor Robot Navigation and Obstacle Avoidance

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

ITA0509-Computer Vision

to the award of the degree of

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**Stereo Vision System for Indoor Robot Navigation and Obstacle Avoidance**" has been carried out by **N.Akshaya(192425129)** under the supervision of **Dr.Yuvaraj S** and is submitted in partial fulfilment of the requirements for the current semester of the **B.Tech Artificial Intelligence And Machine Learning** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **N.Akshaya(192425129)** of the **B.Tech Artificial Intelligence And Machine Learning** ,SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Stereo Vision System for Indoor Robot Navigation and Obstacle Avoidance**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 4/09/2026

Signature of the Students with Names

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Individual Contribution Statement

Student	**Specific Responsibilities**	**Design & Development Contribution**	**Testing & Analysis Contribution**	**Report Contribution**	**Approx. Contribution (%)**
T. Sushma (192325135)	Module 1: Stereo Image Acquisition & Camera Calibration	Stereo image acquisition, camera setup, calibration and preprocessing	Tested camera calibration, image quality and stereo rectification	Module 1 and related methodology sections	**20%**
K. Sai Sahithi (192311240)	Module 2: Depth Estimation & 3D Mapping	Disparity calculation, depth estimation and 3D map generation	Tested depth accuracy and analyzed 3D mapping results	Module 2 and related methodology sections	**20%**
Srinithy.P (192312307)	Module 3: Obstacle Detection & Classification	Obstacle detection and classification using depth information	Tested detection accuracy and classification performance	Module 3 and related methodology sections	**20%**
K. Jaya Sri (192312335)	Module 4: Path Planning & Navigation	Path planning, route generation and indoor navigation	Tested route efficiency and navigation performance	Module 4 and related methodology sections	**20%**
N. Akshaya (192425129)	Module 5: Robot Control & Obstacle Avoidance	Robot movement control and real-time obstacle avoidance	Tested robot movement, collision prevention and avoidance performance	Module 5, testing and conclusion sections	**20%**
Total					**100%**

ABSTRACT

Indoor mobile robots require reliable perception and navigation to move safely through environments containing walls, furniture, people, and other obstacles. Traditional obstacle detection methods may provide limited information about the surrounding environment. Therefore, a vision-based system capable of estimating depth and identifying obstacles can improve indoor robot navigation.

This project presents a **Stereo Vision System for Indoor Robot** for navigation and obstacle avoidance. The proposed system uses two cameras positioned at a fixed baseline to capture left and right images of the environment. The captured images are processed to determine the disparity between corresponding points. The disparity information is then used to estimate the depth of objects in the robot's surroundings.

The system consists of stereo image acquisition, image preprocessing, camera calibration, disparity estimation, depth calculation, obstacle detection, free-space analysis, navigation decision-making, and motor control. The detected environment is analyzed to determine whether the robot can move forward or should change its direction. When an obstacle is detected in the robot's path, the system evaluates the available space and selects a suitable direction for movement.

The proposed system is designed for indoor environments such as classrooms, laboratories, corridors, and rooms. The system is evaluated through tests involving stereo image capture, depth estimation, obstacle detection, navigation, and obstacle avoidance. The expected outcome is an indoor robot capable of using stereo vision information to improve environmental perception and perform safer autonomous navigation.

Keywords: Stereo Vision, Indoor Robot, Depth Estimation, Obstacle Detection, Navigation, Obstacle Avoidance.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	—
CV	Computer Vision	—
RGB	Red Green Blue	—
ROI	Region of Interest	—
FPS	Frames Per Second	—
	Depth	m
	Disparity	pixels
	Focal Length	pixels
	Stereo Baseline	m

CHAPTER 1
INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Indoor mobile robots are designed to perform tasks such as monitoring, inspection, transportation, assistance, and service operations. For autonomous operation, a robot needs to understand its surrounding environment and make suitable movement decisions.

Indoor environments contain different obstacles such as walls, tables, chairs, boxes, equipment, and people. A robot must detect these obstacles and estimate their distance to avoid collisions.

Stereo vision is a computer vision technique that uses two cameras to capture the same scene from different viewpoints. The difference between corresponding points in the two images is called disparity. This disparity can be used to estimate the depth of objects.

The proposed **Stereo Vision System for Indoor Robot** uses stereo cameras to obtain depth information from the environment. This information is used for obstacle detection, free-space identification, navigation, and obstacle avoidance.

1.2 Need for the Project

Indoor robot navigation requires continuous awareness of the surrounding environment. A robot without reliable obstacle detection may collide with objects or select an unsafe path.

The project is required to:

- Detect obstacles in the robot's path.
- Estimate obstacle distance.
- Identify free navigation areas.
- Improve autonomous movement.
- Reduce collision risk.
- Provide depth information.
- Support safe indoor navigation.

1.3 Problem Statement

An indoor robot must navigate safely through environments containing static and moving obstacles. Conventional distance sensors may provide information only from specific directions and may not provide sufficient spatial information about the complete environment.

Therefore, the project aims to develop a stereo vision-based system capable of obtaining depth information, detecting obstacles, identifying free space, and generating navigation commands for an indoor robot.

1.4 Project Aim

To design and develop a **stereo vision-based indoor robot system for navigation and obstacle avoidance** using depth information obtained from stereo cameras.

1.5 Project Objectives

1. To design a stereo vision system for an indoor robot.
2. To capture left and right images of the environment.
3. To calibrate and preprocess stereo images.
4. To calculate disparity between stereo images.
5. To estimate object depth.
6. To detect obstacles.
7. To identify free-space regions.
8. To develop navigation decision logic.
9. To control robot movement.
10. To test the navigation and obstacle avoidance system.

1.6 Scope

The project focuses on indoor navigation using stereo vision. It includes stereo image acquisition, camera calibration, image processing, disparity estimation, depth estimation, obstacle detection, free-space analysis, navigation decisions, and robot movement.

The project does not focus on GPS-based outdoor navigation, large-scale outdoor mapping, or industrial-scale robotic deployment.

1.7 Expected Engineering Outcome

The expected outcome is a functional indoor robot prototype capable of perceiving its environment using stereo cameras, estimating object depth, detecting obstacles, selecting a suitable path, and performing autonomous obstacle avoidance.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focuses on stereo vision, depth estimation, obstacle detection, autonomous robot navigation, and computer vision-based robotic systems.

Existing sensor-based and camera-based approaches were considered to understand their advantages, limitations, and applicability to indoor robot navigation.

2.2 Review of Existing Work

Stereo Vision

Stereo vision uses two cameras to observe the same environment from different viewpoints. The corresponding points between the images are used to calculate disparity and estimate depth.

Ultrasonic-Based Navigation

Ultrasonic sensors measure the distance to nearby objects. They are simple and economical but provide limited information about the complete environment.

LiDAR-Based Navigation

LiDAR provides accurate distance information and can support advanced robot navigation. However, the hardware can be more expensive than basic camera systems.

Monocular Vision

A monocular camera provides visual information using a single viewpoint. However, obtaining reliable depth information is more difficult compared with calibrated stereo cameras.

Computer Vision-Based Navigation

Computer vision allows a robot to analyze its environment using camera images. Stereo vision improves environmental perception by providing depth information in addition to visual information.

2.3 Comparative Analysis

Table 2.1 – Comparison of Existing Navigation Approaches

Approach	**Advantages**	**Limitations**
Ultrasonic	Low cost and simple	Limited spatial information
Infrared	Simple and low power	Limited range
LiDAR	Accurate distance measurement	Higher cost
Monocular camera	Low hardware requirement	Difficult depth estimation
Stereo vision	Visual + depth information	Requires calibration

2.4 Research / Engineering Gap

Low-cost robot navigation systems may use simple distance sensors that provide limited environmental information. Monocular vision provides visual information but has challenges in reliable depth estimation.

Stereo vision provides a practical approach for obtaining both visual and depth information. The main challenge is achieving reliable depth estimation while maintaining suitable processing speed for indoor navigation.

2.5 Proposed Contribution

The proposed system combines stereo image acquisition, depth estimation, obstacle detection, free-space analysis, navigation decision-making, and motor control into a single indoor robot navigation system.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Table 3.1 – Functional and Non-Functional Requirements

Requirement	Type	Target
Stereo image capture	Functional	Continuous
Depth estimation	Functional	Required
Obstacle detection	Functional	Required
Free-space detection	Functional	Required
Navigation decision	Functional	Automatic
Motor control	Functional	Automatic
Reliability	Non-Functional	Stable operation
Safety	Non-Functional	Reduce collision risk
Processing	Non-Functional	Suitable for indoor operation

3.2 Constraints & Applicable Standards

The system is affected by camera calibration, lighting conditions, processing capability, battery capacity, camera placement, and robot speed.

Correct camera alignment is important because errors in camera positioning can affect disparity and depth estimation. The processing system must also provide sufficient computational capability for image processing.

3.3 Design Specifications

Table 3.2 – Design Specifications

Parameter	Requirement	Priority
Stereo cameras	Two cameras	High
Camera baseline	Fixed	High
Depth estimation	Required	High
Obstacle detection	Required	Very High
Navigation	Automatic	High
Motor control	Required	High
Indoor operation	Required	High
Collision avoidance	Required	Very High

3.4 Alternative Solutions & Evaluation

Three main approaches were considered: ultrasonic sensing, LiDAR, and stereo vision.

Ultrasonic sensing provides a low-cost solution but limited spatial information. LiDAR provides accurate depth information but may increase system cost. Stereo vision provides visual and depth information and is selected for the proposed system.

Table 3.3 – Alternative Solution Evaluation

Criterion	Ultrasonic	LiDAR	Stereo Vision
Performance	Medium	High	High
Cost	Low	High	Medium
Depth information	Limited	Excellent	Good

Criterion	Ultrasonic	LiDAR	Stereo Vision
Visual information	Low	Low	High
Hardware complexity	Low	Medium	Medium
Overall suitability	Medium	High	**High**

3.5 Selected Design & Detailed Design

The stereo vision approach is selected because it provides environmental depth information using two cameras while also retaining visual information.

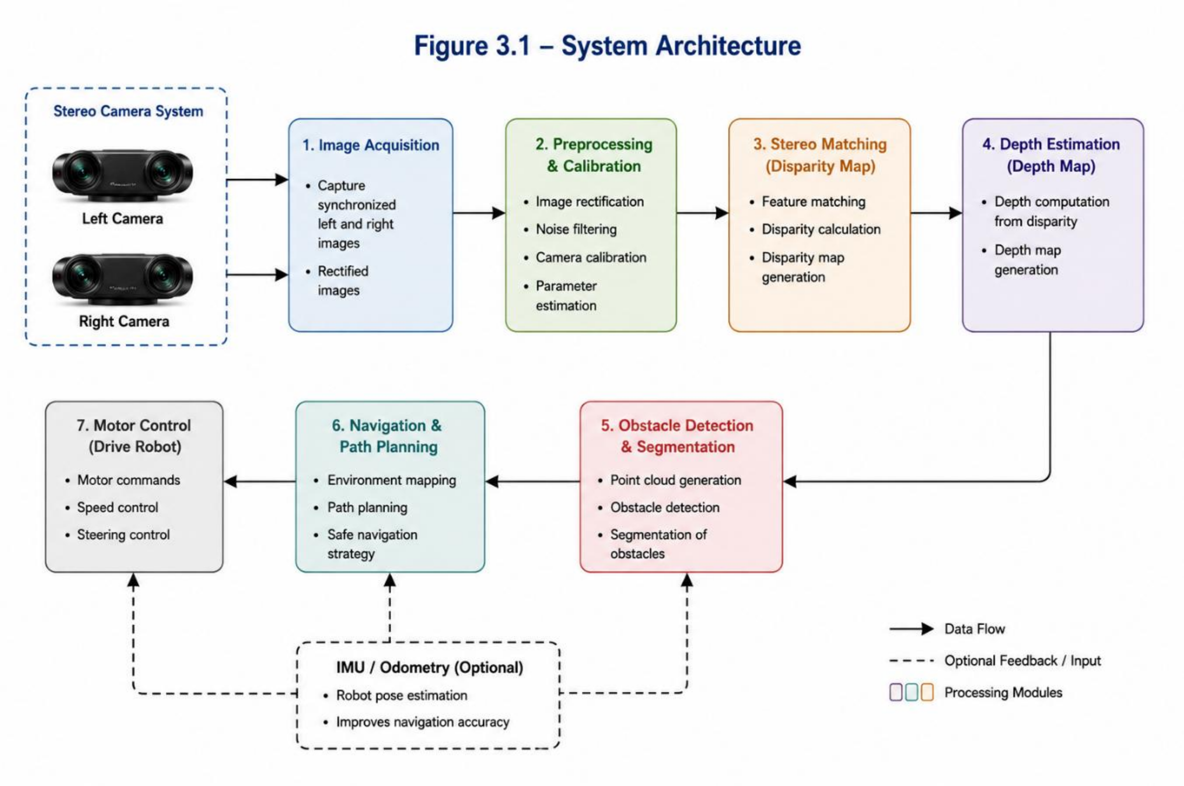

Figure 3.1 – Proposed Stereo Vision System Architecture

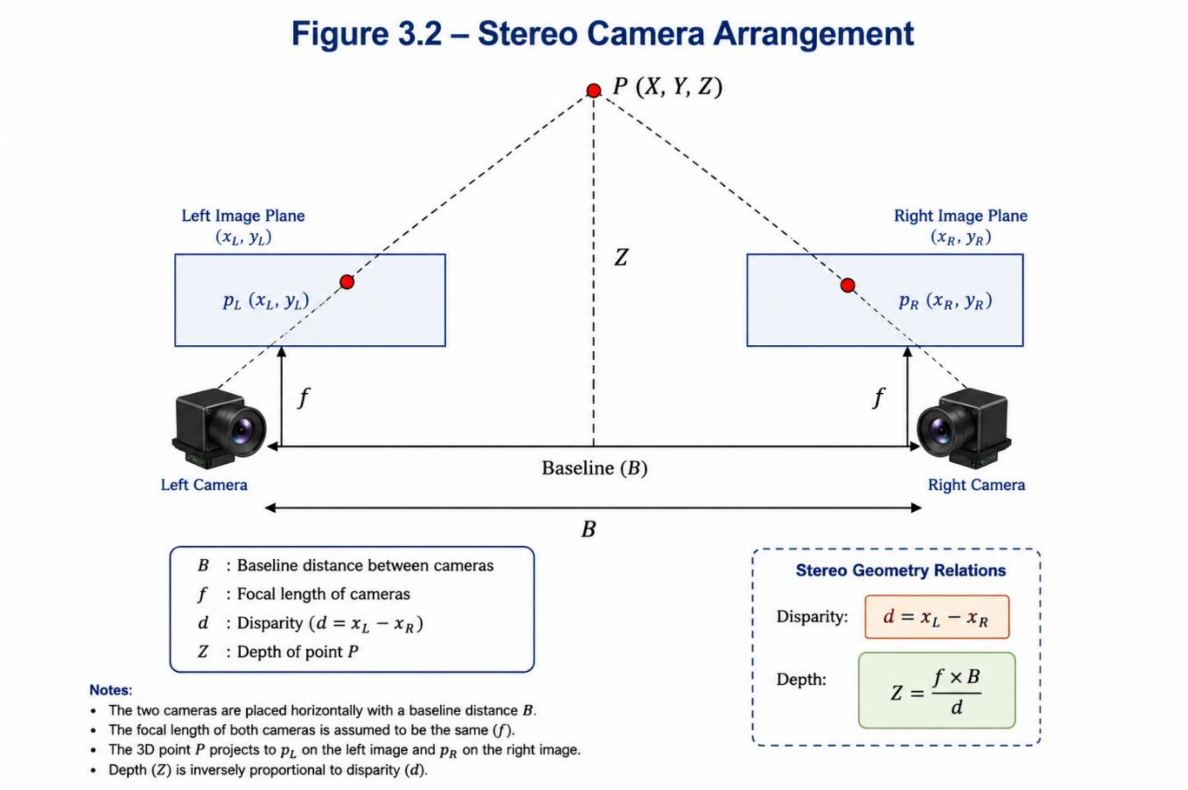

Figure 3.2 – Stereo Camera Arrangement on Indoor Robot

The two cameras are mounted on the front of the robot with a fixed distance between them. Both cameras capture overlapping views of the environment. The difference between corresponding points is used to estimate depth.

Depth Estimation

The basic stereo depth relationship is:

where:

- = depth
- = focal length
- = baseline
- = disparity

Navigation and Obstacle Avoidance

The navigation system continuously analyzes the depth information obtained from the stereo cameras.

The environment is divided into three main regions:

- Left
- Center
- Right

If the center path is clear, the robot moves forward. If an obstacle is detected in the center, the system evaluates the left and right regions.

The robot then selects a safer direction based on available free space.

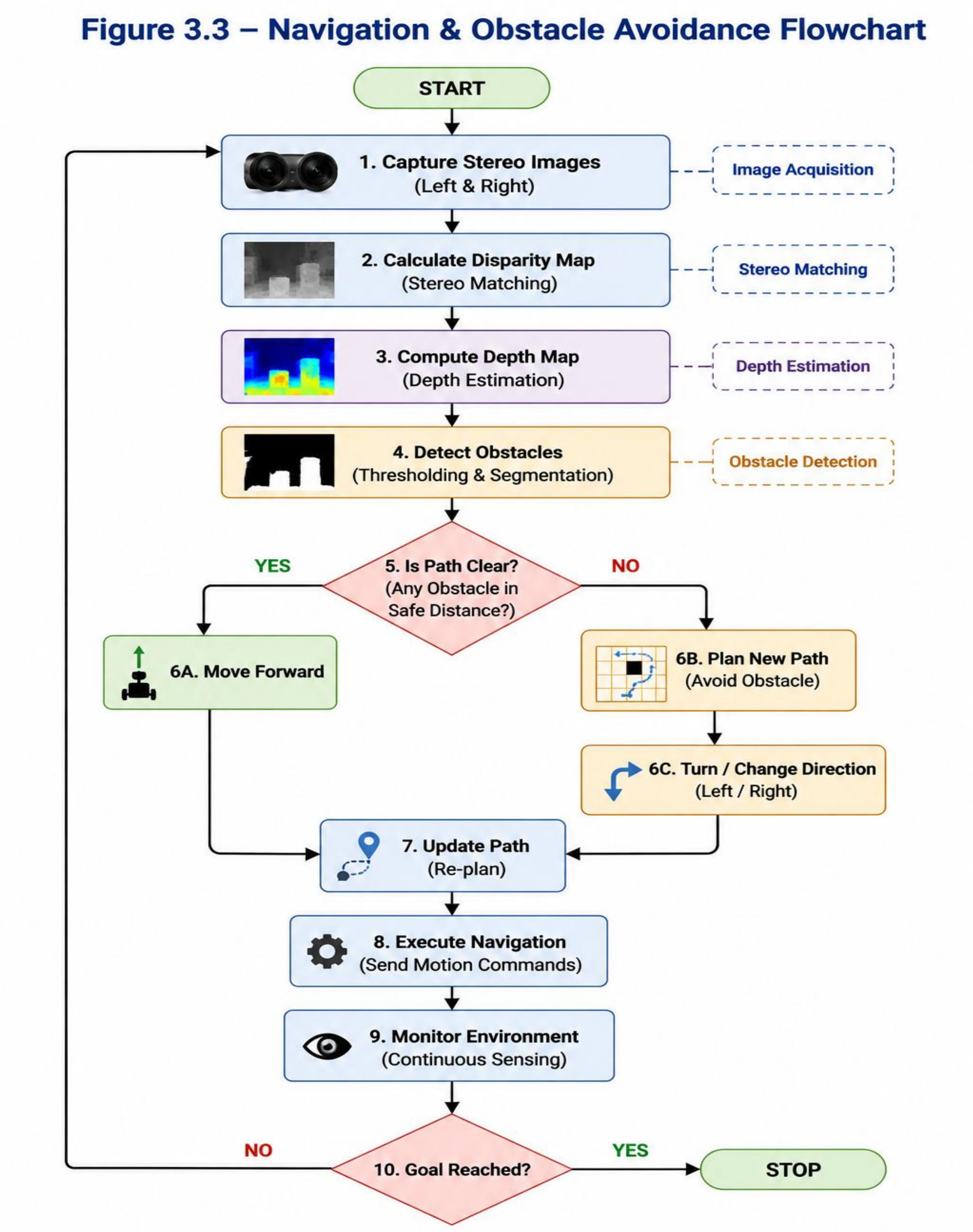

Figure 3.3 – Navigation and Obstacle Avoidance Flowchart

3.6 Trade-Off Analysis

Stereo vision was selected instead of only using proximity sensors because it provides both visual and depth information. The main disadvantage is the additional processing requirement and the need for camera calibration.

The final design therefore balances environmental perception, system cost, computational requirements, and navigation safety.

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability

The system uses reusable electronic components and software-based perception. Stereo cameras provide visual and depth information through a single integrated sensing arrangement.

Ethics

Camera data should be handled responsibly. Images containing people should not be unnecessarily stored or distributed.

Health & Safety

Robot movement should be controlled during testing. The robot should stop when an obstacle is too close or when a safe path cannot be identified.

Security

If images or sensor information are stored or transmitted, access to the information should be appropriately controlled.

Table 3.4 – Risk Register

Risk	Likelihood	Severity	Mitigation	Residual Risk
Poor lighting	Medium	Medium	Improve lighting	Low
Camera misalignment	Medium	High	Calibration	Low
Incorrect depth	Medium	High	Filtering/calibration	Medium
Moving obstacle	Medium	High	Continuous detection	Medium
Processing delay	Medium	Medium	Optimize processing	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system is developed in stages. First, the stereo cameras are mounted and calibrated. The cameras capture left and right images of the indoor environment. The images are processed to calculate disparity and generate depth information.

The depth information is then analyzed to detect obstacles and determine free-space regions. The navigation module uses this information to select a movement direction and provide commands to the robot motors.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

The hardware consists of a robot chassis, stereo cameras, processing unit, motors, motor driver, battery, and connecting components.

The software performs image acquisition, preprocessing, stereo matching, disparity calculation, depth estimation, obstacle detection, navigation, and motor control.

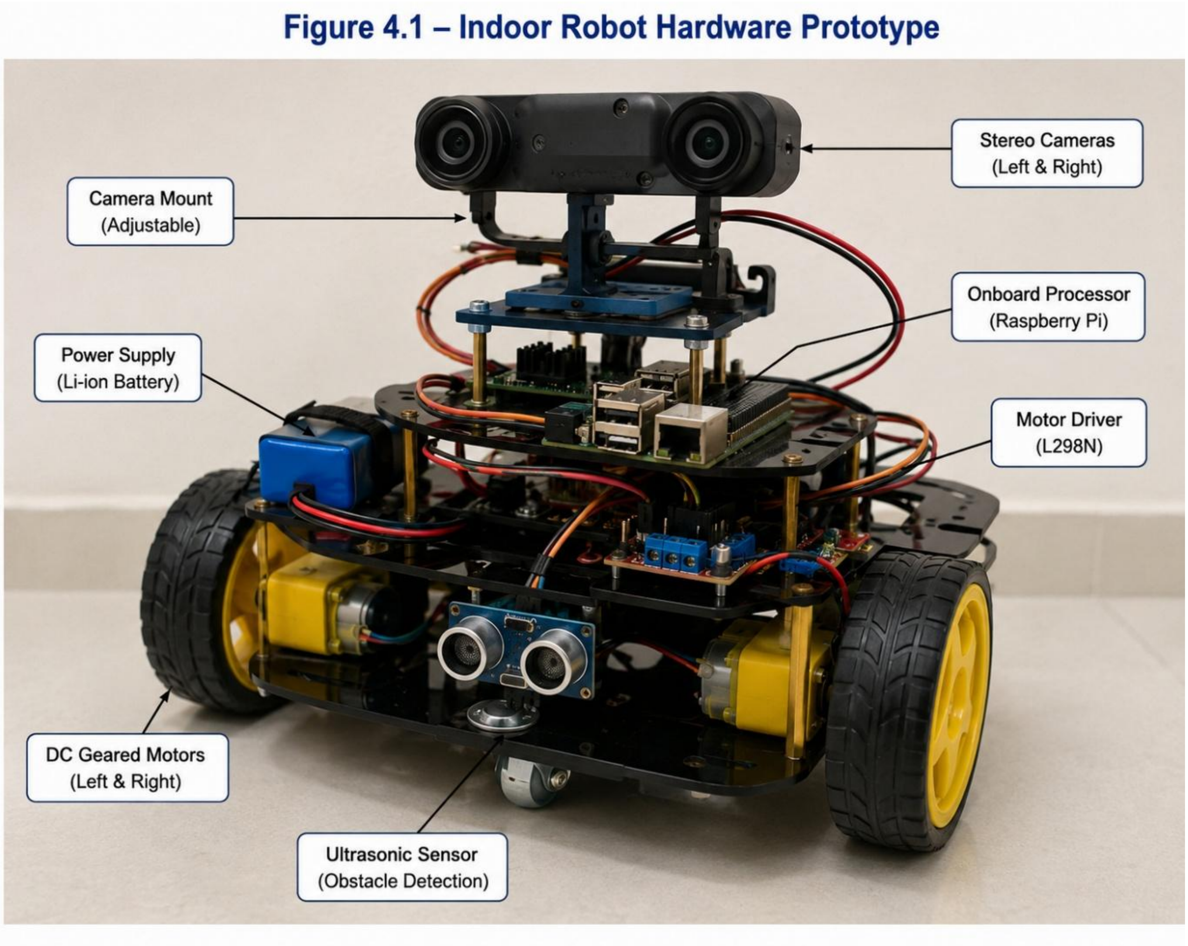

Figure 4.1 – Indoor Robot Hardware Prototype

This figure should show your **actual robot prototype**, including the stereo camera arrangement and major hardware components.

4.3 Test Plan & Verification

Table 4.1 – Test Plan and Verification

Requirement	**Test Method**	**Acceptance Criterion**
Stereo image capture	Capture left/right images	Both images obtained
Camera calibration	Calibration test	Correct alignment
Depth estimation	Objects at known distances	Approximate depth obtained
Obstacle detection	Place obstacle in path	Obstacle detected
Free-space detection	Test open/blocked areas	Safe region identified
Navigation	Place obstacles at different positions	Correct direction selected
Collision avoidance	Place obstacle ahead	Robot stops/changes direction

4.4 Results

The stereo cameras capture two views of the indoor environment. These images are processed to obtain disparity information.

The disparity information is converted into depth information. The resulting depth information allows the system to identify objects that are closer to the robot.

When an obstacle is detected in the forward navigation region, the system checks the available space on the left and right sides and selects an appropriate navigation direction.

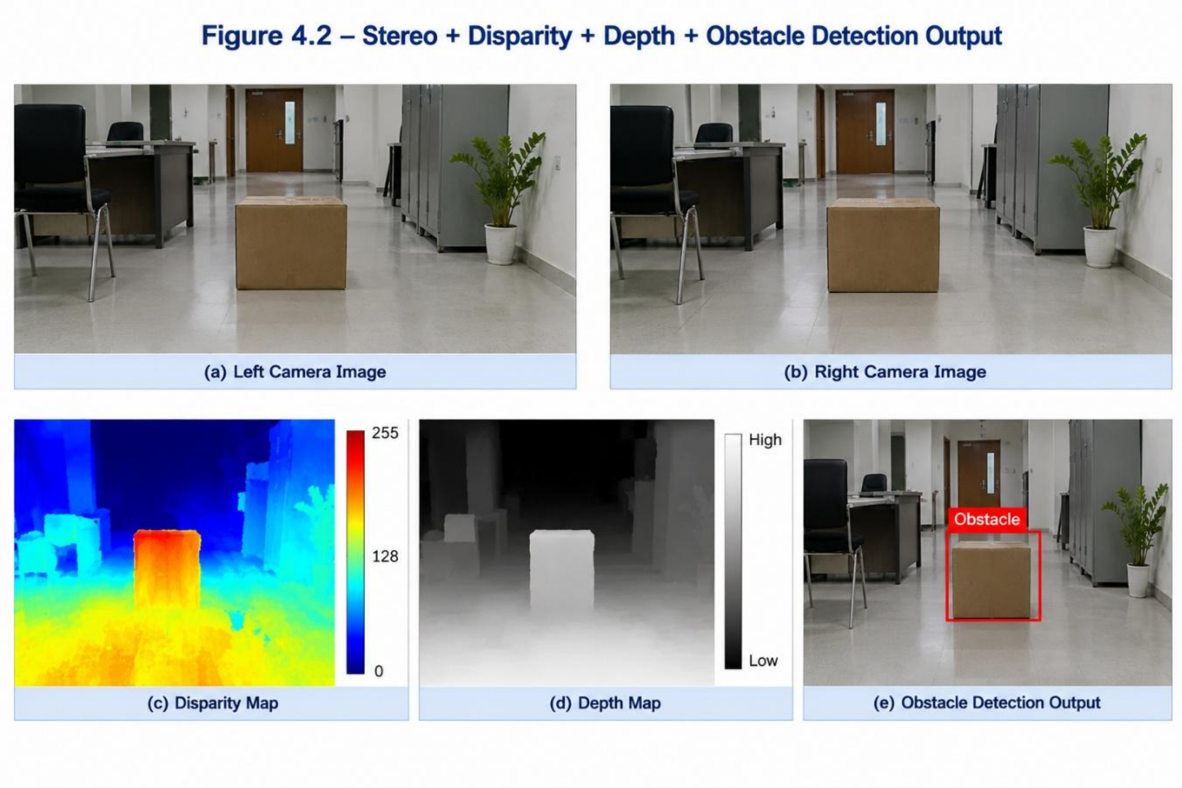

Figure 4.2 – Stereo Vision and Obstacle Detection Output

This **single figure should contain 3–4 panels**:

- **(a)** Left and right stereo images
- **(b)** Disparity map
- **(c)** Depth map
- **(d)** Obstacle detection/navigation output

This keeps the entire report to **only 5 figures**.

4.5 Data Analysis & Engineering Judgment

The system can be evaluated using depth estimation error, obstacle detection performance, navigation success rate, and response time.

Depth error is calculated as:

Percentage error is:

The performance can be affected by lighting conditions, camera calibration, object texture, camera baseline, processing capability, and robot movement.

Actual experimental values should be entered after testing.

4.6 Comparison with Existing Solutions & Achievement of Objectives

The proposed system provides combined visual and depth information for indoor navigation. It is intended to provide more environmental information than simple single-point obstacle sensors.

The achievement of each objective should be verified using the corresponding prototype output and test results.

4.7 Strengths & Limitations

Strengths

- Provides visual and depth information.
- Supports autonomous indoor navigation.
- Detects obstacles in the robot's path.
- Supports safer movement decisions.
- Uses camera-based environmental perception.
- Suitable for indoor environments.

Limitations

- Performance depends on lighting.
- Accurate camera calibration is required.
- Textureless surfaces may reduce stereo matching quality.
- Depth processing requires computational resources.
- Dynamic obstacles may require advanced tracking.

4.8 Project Planning, Roles & Budget

Project Planning

Phase	Activity
1	Problem identification
2	Literature review
3	Hardware selection
4	Stereo camera setup
5	Software development

CHAPTER 5
CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The **Stereo Vision System for Indoor Robot** was designed to provide depth-based environmental perception for navigation and obstacle avoidance. The proposed system combines stereo image acquisition, image preprocessing, camera calibration, disparity estimation, depth calculation, obstacle detection, free-space analysis, navigation decision-making, and motor control.

The stereo vision approach enables the robot to obtain spatial information about objects in its environment. This information can be used to identify obstacles and select an appropriate movement direction.

The system provides a practical foundation for autonomous indoor robot navigation. Its performance is influenced by factors such as camera calibration, lighting conditions, image quality, processing capability, and environmental conditions.

5.2 Future Work

Future improvements can include:

- Advanced stereo matching algorithms.
- Deep-learning-based obstacle detection.
- Dynamic obstacle tracking.
- SLAM-based indoor mapping.
- Advanced path planning.
- LiDAR and stereo vision integration.
- Improved low-light performance.
- Real-time processing optimization.
- Mobile application for robot monitoring.
- Multi-room autonomous navigation.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Stereo vision principles.
- Camera calibration.
- Image processing.
- Disparity estimation.

- Depth estimation.
- Obstacle detection.
- Robot navigation.
- Motor control.

Engineering & Professional Skills Developed

- Computer vision implementation.
- Hardware integration.
- Software development.
- Robot control.
- Testing and debugging.
- Data analysis.
- Engineering problem solving.
- Technical documentation.
- Teamwork.

Individual Reflection

The project provided practical experience in integrating computer vision with robotic navigation. The major challenge was understanding stereo disparity and converting it into useful depth information for navigation. Testing helped in understanding the effects of camera alignment, lighting, and obstacle position on system performance. The project improved problem-solving, debugging, teamwork, and technical documentation skills. If the system were redesigned, more attention would be given to real-time processing and dynamic obstacle handling.

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APPENDICES

Appendix A – Detailed Calculations

This appendix contains the important calculations used in the stereo vision system.

- Stereo depth calculation.
- Disparity calculation.
- Depth error calculation.
- Percentage error calculation.
- Navigation-related calculations.

Appendix B – Engineering Drawings / Schematics

This appendix contains:

- Stereo camera arrangement.
- Robot system architecture.
- Robot wiring diagram.
- Motor driver connections.
- Navigation flowchart.

Appendix C – Source Code / Code Repository Information

This appendix contains essential code sections for:

- Stereo image acquisition.
- Camera calibration.
- Disparity calculation.
- Depth estimation.
- Obstacle detection.
- Navigation and motor control.

Appendix D – Datasheets

Datasheets for:

- Stereo camera.
- Processing board.
- Motor driver.
- Motors.
- Battery.

Appendix E – Experimental / Raw Data

This appendix contains:

- Stereo image samples.
- Disparity outputs.
- Depth measurements.
- Obstacle detection observations.
- Navigation test observations.

Appendix F – Ethics / Institutional Approval

This appendix contains relevant:

- Camera privacy considerations.
- Indoor testing safety considerations.
- Data handling considerations.

Appendix G – Project Meeting / Progress Records

- Project meeting records.
- Development progress.
- Testing progress.
- Mentor feedback.

Appendix H – User / Industry Feedback

- User feedback.
- Mentor feedback.
- Demonstration feedback.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Include this section only if applicable:

- Publications.
- Competitions.
- Demonstrations.
- Prototype outcomes.

Appendix J – Individual Contribution Evidence

N.Akshaya(192425129)

Module 5: Robot Control & Obstacle Avoidance

Module 5 focuses on implementing the **robot control and obstacle avoidance component** of the stereo vision-based indoor navigation system. The module begins with the depth and obstacle information generated by the previous vision-processing stages. This information is used to determine the distance of obstacles from the robot and identify whether the current navigation path is safe.

The input depth map and obstacle detection results are first processed to identify **free-space regions, obstacle locations, and safe navigation directions**. The system continuously analyzes the forward region of the robot and compares the estimated obstacle distance with a predefined safety threshold. If no obstacle is detected within the safe distance, the robot is allowed to move forward along the planned path.

When an obstacle is detected in the robot's path, the module evaluates the **left and right free-space regions** to determine a suitable alternative direction. Based on the available space and navigation requirements, the robot selects a safer direction and generates appropriate movement commands such as **forward, left turn, right turn, or stop**. This enables the robot to avoid obstacles while maintaining safe indoor navigation.

The generated navigation decision is converted into **motor control commands** and transmitted to the robot's motor driver. The motor controller regulates the movement and direction of the left and right motors. The system continuously receives updated stereo-vision information and repeats the obstacle detection and navigation decision process to respond to changing indoor environments.

The module evaluates the effectiveness of the control and avoidance mechanism using parameters such as **obstacle detection distance, collision avoidance success, navigation accuracy, response time, path-following performance, and successful obstacle avoidance rate**. The resulting control actions enable the robot to navigate safely and autonomously through indoor environments.

Main Functions:

· Depth and obstacle information processing
· Free-space identification
· Obstacle distance analysis
· Safety-distance threshold checking
· Forward-path clearance evaluation
· Left and right free-space analysis
· Safe-direction selection
· Navigation decision generation
· Obstacle avoidance decision making

· Forward, left, and right movement control

· Stop command generation for unsafe conditions

· Motor driver control

· Continuous environment monitoring

· Real-time navigation and obstacle avoidance

· Collision-risk reduction

· Path replanning after obstacle detection

· Evaluation of obstacle avoidance and robot control performance